

Two Faces of Plasticity in Small Spiking Circuits: How Connectome Topology and Cell-Type-Specific Learning Rules Shape What a Small Network Can Learn

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Abstract

A nervous system is not a static wire diagram: it is shaped by plasticity at every level, from the local rule a single synapse obeys to the global topology that a lifetime of learning leaves behind. We study two complementary faces of this organization in small, fully reproducible spiking-network models inspired by the nematode *C. elegans*. *Project I* asks a **structural** question: does the heavy-tailed weight topology that biology universally expresses encode a *memory of past learning* that biases future learning, and do different starting topologies converge under sustained training? Using a 300-neuron leaky integrate-and-fire (LIF) recurrent network trained by eligibility propagation (e-prop) on a sequence of seven behavioural tasks, we show that a heavy-tailed connectome trades long-term flexibility for early-task speed: it acquires new tasks slightly faster at first but forgets more, while a uniform connectome gradually *grows* a heavy tail. Across complementary statistics (Gini coefficient, top-5% weight share, power-law exponent), both topologies migrate toward a shared heavy-tail basin whose location is set jointly by network structure and task demand. *Project II* asks a *rule-level* question: does assigning **different** plasticity rules to different cell types beat a single shared rule? In a 34-cell LIF model of chemotaxis, heterogeneous rule assignment decisively outperforms a strong homogeneous baseline under real-time spike readout (Endpoint +0.84 vs. -0.64, all contrasts $p < 10^{-7}$), because a single reward-driven rule saturates every layer and flattens the concentration-to-value gradient the worm turns by; distributing rules keeps that gradient alive. Together, the two projects argue that *how plasticity is organized*—in the structure it slowly inscribes, and across the cell types it is divided among—is as decisive for small-network computation as the plasticity itself.

Keywords: synaptic plasticity; spiking neural networks; *C. elegans*; connectome topology; heavy-tailed synaptic weights; eligibility propagation (e-prop); three-factor Hebbian rules; cell-type-specific plasticity; BCM; chemotaxis; network saturation.

1 Introduction

A defining property of nervous systems is that they change with experience. The strength of a synapse is not fixed: it is adjusted by local biochemical machinery that detects coincidences of pre- and post-synaptic activity, by global neuromodulatory signals that report reward or arousal, and by homeostatic processes that keep firing within a viable range (Bi & Poo, 2001; Frémaux & Gerstner, 2016; Hebb, 1949; Turrigiano, 2012). Out of these local events emerges something global—a behavioural repertoire, a memory, a self. Understanding how the *organization* of plasticity produces this emergence is a central problem of computational neuroscience.

The nematode *C. elegans* is an unusually clean stage on which to pose this problem. Its nervous system is compact (302 neurons) and fully mapped at the level of chemical synapses and gap junctions (Cook et al., 2019; White et al., 1986), yet its cells fall into more than a hundred

transcriptionally and morphologically distinct classes, each expressing a particular complement of ion channels, neurotransmitters, receptors and plasticity-related molecules (Bargmann, 2006; Marder & Goaillard, 2006). The forms of plasticity a neuron can express are correspondingly diverse: spike-timing-dependent Hebbian rules (Bi & Poo, 1998; Song et al., 2000), sliding-threshold selectivity rules such as BCM (Bienenstock et al., 1982), reward- or neuromodulator-gated “three-factor” rules (Frémaux & Gerstner, 2016; Gerstner et al., 2018; Izhikevich, 2007), and homeostatic stabilization (Turrigiano et al., 1998; Turrigiano, 2012). In parallel, the *statistics* of the connectome that emerges from a life of such plasticity are remarkably consistent across species: synaptic weights are heavy-tailed, with a few very strong connections amid many weak ones (Buzsáki & Mizuseki, 2014; Schröter et al., 2017).

These two observations motivate two complementary questions, which we study in two projects.

Q1 (structural; Project I). Because a heavy-tailed weight distribution is what training *produces* in artificial networks and what biology *displays*, we ask whether a heavy-tailed topology should be regarded as a compact *record of past learning*, and whether this record biases the network’s ability to learn further. If two otherwise identical networks—one starting heavy-tailed, one starting uniform—are trained on a long sequence of tasks, do they learn differently, and do their topologies converge to a common attractor?

Q2 (rule-level; Project II). Because plasticity is molecularly diverse and cell-type-specific, we ask whether *distributing* different plasticity rules across cell types (heterogeneous assignment) confers a computational advantage over applying a single shared rule everywhere (homogeneous assignment)—and, if so, whether the advantage is causal (the specific rule→type pairing) or merely a consequence of having some diversity.

Both questions are hard to answer cleanly. An advantage of a topology or a rule assignment can be hidden by a permissive downstream readout; and distinguishing “diversity matters” from “the specific mapping matters” demands a particular ablation (randomly reassigning the same rules) that is easy to omit. We control for both in deliberately small, fully specified models.

Contributions.

1. **Project I:** A 300-neuron LIF recurrent network, trained with e-prop (Bellec et al., 2020) under continuous (non-reset) learning across seven behavioural tasks, lets us compare heavy-tailed and uniform connectomes. We find that the heavy-tailed topology speeds early acquisition but accelerates forgetting, and that both topologies drift toward a shared heavy-tail basin—evidence that a heavy tail is an attractor set by structure and task demand rather than an arbitrary initial condition.
2. **Project II:** A 34-cell LIF chemotaxis model with five cell types and five plasticity rules, evaluated under two readouts (ideal cached lookup vs. continuous real-time spikes), shows that heterogeneous rule assignment beats a strong homogeneous baseline specifically under real-time computation, by avoiding network-wide saturation that would otherwise flatten the chemotactic gradient.
3. We unify the two into a single message: *plasticity pays when it is organized*—whether that organization is the slow inscription of a heavy-tailed topology, or the partition of rules across cell types—and we discuss the limitations and the natural next experiments.

2 Project I: Heavy-Tailed Connectome Topology as a Memory of Past Learning

2.1 Hypothesis and background

When artificial neural networks are trained on realistic tasks, their weight matrices develop a heavy-tailed distribution: most weights stay small, but a few grow large (Buzsáki & Mizuseki, 2014; Schröter et al., 2017). This is also the near-universal signature of biological connectomes, from *C. elegans* to primate cortex. We advance a simple working hypothesis:

A heavy-tailed weight distribution is the structural residue of accumulated past learning. When such a network meets an entirely new task, the residue acts approximately like a randomly drawn heavy-tailed prior on the weights: it biases, but does not determine, what the network can learn next.

This yields two testable predictions. First, a network that *starts* heavy-tailed (as if it had already learned a great deal) should behave differently on a new task than one that starts uniform—faster early on, perhaps, but more rigid. Second, because the heavy tail is the signature of “having learned,” a uniform network should *grow* one under sustained training, and the two topologies should drift toward each other. We test both.

2.2 Model

We simulate pathfinding on a 3×3 grid by a recurrent network of $N = 300$ current-based LIF neurons, with 270 excitatory and 30 inhibitory units respecting Dale’s principle (inhibition markedly improves learnability at this scale). Sensory input reports the grid state through dedicated channels; four output neurons encode the four movement directions and are read out as a population vector:

$$\hat{\mathbf{d}} = \frac{\sum_k J_k \mathbf{e}_k}{\|\sum_k J_k \mathbf{e}_k\|}, \quad \mathbf{e}_k \in \{(0, 1), (0, -1), (1, 0), (-1, 0)\},$$

where J_k is the spike count of output neuron k over the evaluation window. Membrane potentials obey

$$\tau_m \dot{u}_j = -u_j + \sum_k W_{jk} s_k(t) + I_{\text{ext},j} + \xi_j(t), \quad (1)$$

with filtered pre-synaptic traces $s_k(\tau_s)$, threshold-and-reset spiking, and a refractory period. We compare two *initial connectomes* (Fig. 1): **uniform**, drawn from a narrow, near-homogeneous distribution (Gini ≈ 0.33); and **preferential** (heavy-tailed), generated by preferential-growth wiring that yields a heavy-tailed weight distribution (Gini ≈ 0.73).

2.3 From three-factor Hebbian rules to eligibility propagation

We first trained both connectomes with a biologically local three-factor Hebbian rule: an eligibility trace $e_{ij} = x_{\text{pre},i} x_{\text{post},j}$ that is gated by a global, reward-modulated third factor. In expectation, such rules implement policy-gradient credit assignment locally (Frémaux et al., 2010),

$$\mathbb{E}[\Delta W_{ij}] \propto \frac{\partial J}{\partial W_{ij}},$$

which two-factor Hebbian rules cannot. In practice, however, convergence held only in expectation: on our small network the rule barely exceeded chance on the food-approach task (≈ 0.13 accuracy, Fig. 2), because the expectation is approached only through extensive trial-and-error that a 300-neuron model cannot afford.

We therefore adopted **eligibility propagation (e-prop)** (Bellec et al., 2020), a biologically plausible approximation to backpropagation through time. e-prop replaces the non-differentiable spike with a surrogate derivative $\psi_j = \sigma'(\beta(u_j - V_{\text{th}}))$, filters it into a post-synaptic eligibility

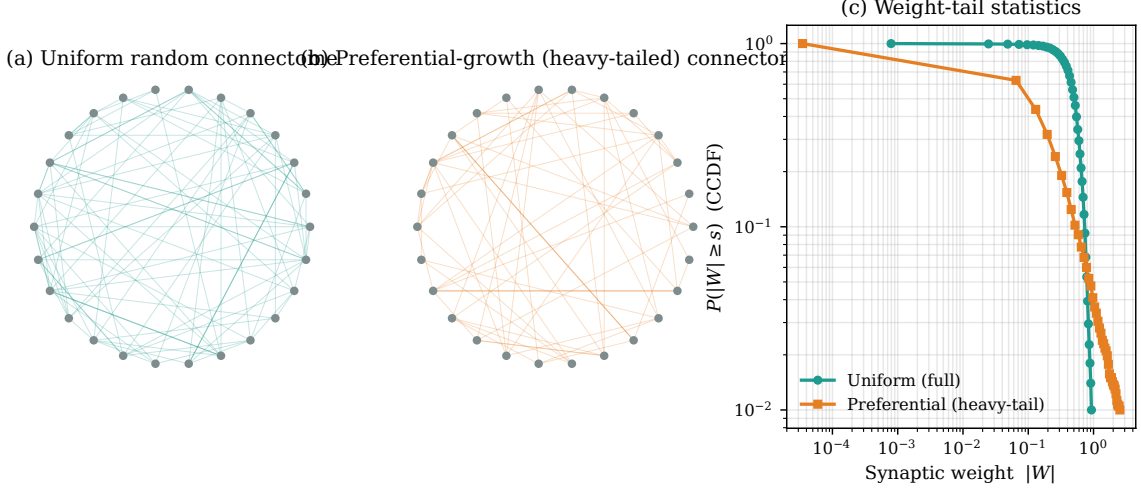


Figure 1. The two starting connectomes. (a) A uniform random connectome with a near-homogeneous weight distribution. (b) A preferential-growth connectome with a heavy-tailed weight distribution, mirroring the statistics of biological connectomes (Buzsáki & Mizuseki, 2014; Schröter et al., 2017). (c) Complementary cumulative distribution $P(|W| \geq s)$ of synaptic strengths: the heavy-tailed topology carries a much fatter tail (slower decay) than the uniform one.

trace p_j ($\tau_e \dot{p}_j = -p_j + \psi_j$), and forms the synaptic eligibility trace as the product of pre- and post-synaptic traces:

$$e_{jk}(t) = p_j(t) \cdot s_k(t). \quad (2)$$

Crucially, the learning signal is *not* back-propagated through the forward weights. Instead it is generated by feedback alignment (Lillicrap et al., 2016): a fixed random feedback matrix B broadcasts the readout error δ_{out} to each neuron,

$$L_j(t) = (B \delta_{\text{out}}(t))_j, \quad \Delta W_{jk} = -\eta_{\text{rec}} \sum_t L_j(t) e_{jk}(t). \quad (3)$$

Because B need not be symmetric to W , this sidesteps the weight-transport problem while retaining per-neuron (rather than network-global) credit assignment. With e-prop, both connectomes solved the single food-approach task to ≈ 0.89 accuracy (Fig. 2). The biological reading of Eq. (2) is appealing: the pre-synaptic trace s_k resembles a post-synaptic potential, while the surrogate term ψ_j maps onto NMDA- Ca^{2+} coincidence detection and the slow Ca^{2+} tag of a dendritic spine—a candidate substrate for the eligibility trace (Gerstner et al., 2018).

2.4 Single-task indistinguishability motivates a multi-task design

Under e-prop on a *single* task, the two connectomes performed almost identically (Fig. 3): when each of the seven tasks is trained from scratch, the uniform and heavy-tailed connectomes converge to the same final accuracy and reach the learning threshold at nearly the same trial. e-prop’s credit assignment is powerful enough to override any difference the starting topology makes on one isolated problem. This null result is itself informative—it implies that topology matters only when the network is stressed by *many* tasks. We therefore designed a sequential multi-task protocol of seven behaviours, chosen to span the ethological repertoire of a small animal: (i) approach food; (ii) flee a predator; (iii) freeze (“play dead”) under a large threat; (iv) detour around a conspecific; (v) avoid an experimental shock source; (vi) carry food or home, gated by a context bit (carry $\rightarrow$ home, otherwise $\rightarrow$ food); and (vii) moisture seeking, also context-gated (dry $\rightarrow$ approach wet, wet $\rightarrow$ avoid dry). Tasks (vi)–(vii) force the recurrent network to perform genuine gating computation, which is the regime most sensitive to topology. Critically, learning is *continuous*: the recurrent weights W and the shared readout W_{out} are never reset

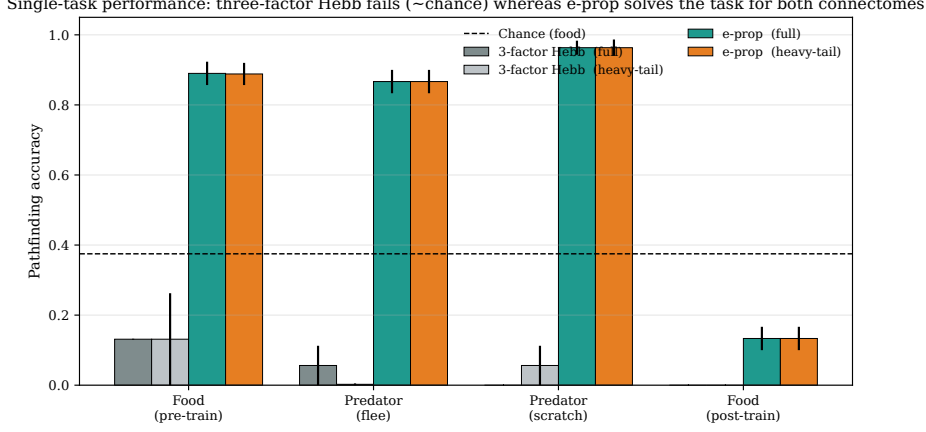


Figure 2. Learning-rule selection. Three-factor Hebbian plasticity (grey) barely exceeds chance on single tasks, whereas e-prop (teal/orange) solves them for *both* connectomes. We therefore adopt e-prop for all subsequent experiments. Error bars: s.d. over seeds.

Table 1. Three-factor Hebbian plasticity versus e-prop. e-prop is essentially a credit-assignment upgrade of the three-factor rule: it keeps the biologically local, per-neuron character while replacing the single broadcast reward with a per-neuron learning signal carried by random feedback.

Dimension	Three-factor Hebbian	e-prop
Eligibility trace	$x_{\text{pre}} \cdot x_{\text{post}}$ (coarse)	$p_j \cdot s_k$, with surrogate-gradient dynamics
Third factor	scalar reward ($M - b$), broadcast network-wide	per-neuron $L_j = B \delta_{\text{out}}$
Feedback	none (scalar only)	random feedback weights B (feedback alignment)
Target information	only a “score”	supervised food direction y^*
Credit assignment	very coarse	fine (one channel per neuron)
Sample efficiency	low (≈ 0.5 over ~ 2000 trials)	high (≈ 1.0 within a few hundred trials)
Biological realism	strong (reward-modulated local Hebb)	fairly strong (local traces + random feedback; no weight symmetry)

between tasks, so later tasks are learned on top of the structure that earlier tasks left behind. Each task was trained for 200 trials and evaluated for 200 trials, over three random seeds.

2.5 Results: acquisition, retention, and forgetting

At acquisition (Fig. 4a), the two connectomes are nearly indistinguishable: the uniform network is marginally but consistently ahead on the harder tasks (detour, shock-avoidance). The decisive separation appears at retention (Fig. 4b): when every task is re-evaluated on the *final* connectome, the uniform network remembers far more. Quantitatively, catastrophic forgetting (acquisition minus retention, Fig. 5) is markedly larger for the heavy-tailed network on the demanding tasks: on “play dead,” the heavy-tailed network forgets 0.64 of its acquired accuracy versus 0.15 for the uniform network, and on “carry/home” it forgets 0.56 versus 0.50.

The picture so far is a *final-state* summary: across tasks the two connectomes acquire similar accuracy, but the heavy-tailed one forgets more. Reading the per-trial learning curves, however, reveals *when* each topology holds the upper hand—and the answer is the key to the entire trade-off, which we unpack next (Section 2.6).

Single-task learning curves: both connectomes converge to similar accuracy (topology is indistinguishable in isolation)

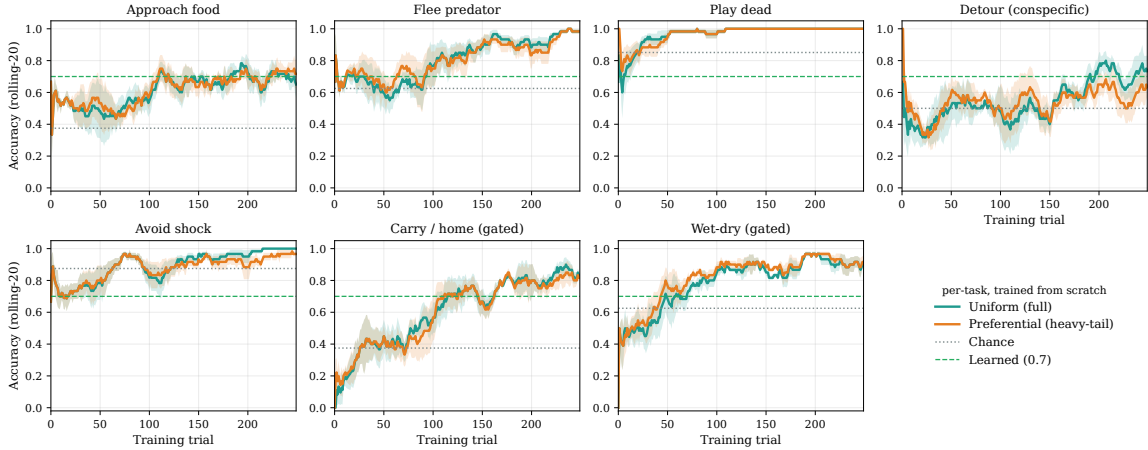


Figure 3. Single-task learning curves (each task trained from scratch). Under e-prop, the uniform (teal) and heavy-tailed (orange) connectomes learn each task to the same final accuracy and reach the learning threshold (dashed green, 0.7) at nearly the same trial. Shaded bands: s.e.m. over three seeds; dotted grey: per-task chance level. Topology is indistinguishable in isolation—which is precisely why we move to continuous multi-task learning, where the residue of past learning can bias what comes next.

2.6 The expert–novice trade-off: why the prior helps early but hurts late

The continuous-training learning curves (Fig. 6) expose a crossover that the final-state averages concealed. In the early phase of nearly every task—roughly the first 30 trials, shaded—the heavy-tailed connectome is ahead: its rolling accuracy climbs faster and sits above the uniform one. Late in training the order reverses, and the uniform connectome finishes at higher accuracy. Aggregated across the seven tasks, the heavy-tailed network’s mean accuracy advantage is $+0.017$ in the early phase (first 30 trials) and -0.019 in the late phase (last 30 trials) (Fig. 6h): modest in absolute size, but a consistent sign change, and it is amplified on the demanding context-gated tasks—on *carry/home* the expert’s early edge reaches $+0.10$. We now explain each side of the trade-off.

Why the expert starts ahead. A heavy-tailed connectome carries a few very strong recurrent connections—*hubs*—amid many weak ones. Before any learning on the new task has occurred, these hubs already endow the recurrent dynamics with structured motifs (sustained activity, integration, directional selectivity) that a uniform matrix does not. When a sensory input arrives, the strong pathways propagate it reliably to the output, so the network produces a non-trivial, often-correct behavioural response from the very first trial; the uniform network’s activity is more diffuse and stays near chance until learning builds the pathways. The hubs also raise the effective gain and signal-to-noise of the input-to-output mapping, and—because they dominate the dynamics—they focus the e-prop eligibility traces and learning signal on the synapses that matter, so the first weight updates are large and coherent. In short, the strong connections are a reusable *scaffold*: plasticity borrows them to erect a workable initial policy. This is the macro-level face of our hypothesis—the heavy tail is the residue of past learning, and that residue is a useful prior on a new task.

Why the novice finishes ahead. The same scaffold that accelerates early learning becomes a liability once the network must solve the task *optimally* and *stably*. First, the expert’s strong connections bias the optimisation: they lock the dynamics into patterns inherited from the prior, so e-prop converges to a basin shaped by those hubs rather than the basin best suited to

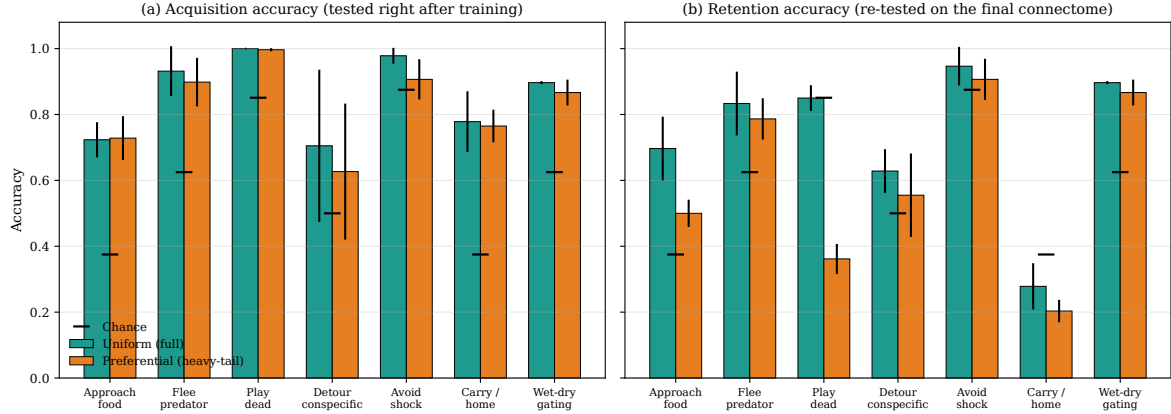


Figure 4. Multi-task performance. (a) Acquisition accuracy, tested immediately after each task is trained. The two connectomes are closely matched, with the uniform network marginally ahead on most tasks. (b) Retention accuracy, re-tested on the *final* connectome after all seven tasks. The uniform network retains substantially more—the gap widens on the most demanding tasks (play-dead, carry/home). Black ticks mark per-task chance level. Error bars: s.d. over seeds.

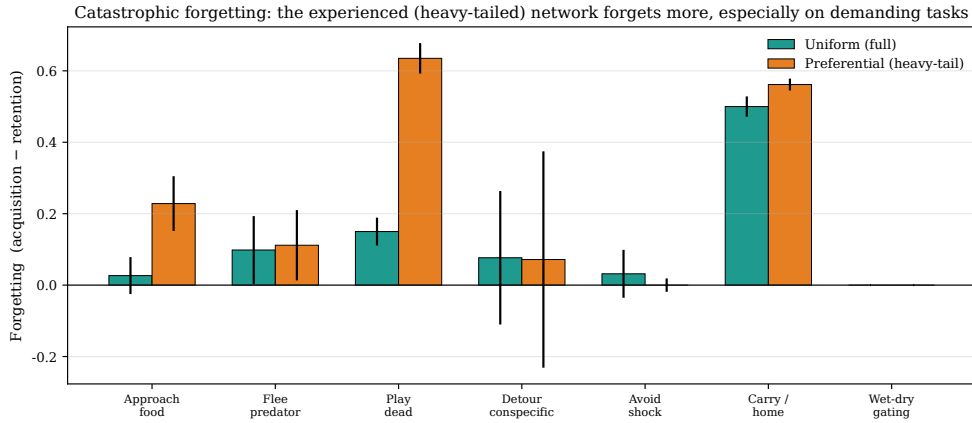


Figure 5. Catastrophic forgetting. The heavy-tailed (“experienced”) network forgets more than the uniform (“naïve”) network, most strikingly on play-dead and carry/home. The few strong connections that encode the heavy tail appear to be overwritten preferentially when a new, demanding task arrives.

the current tasks. The novice, starting from an unbiased, flexible substrate, is free to find the better basin. Second, the expert’s few dominant synapses are hard to move—they sit near the operating point and control a great deal of the dynamics at once—so the network plateaus at lower accuracy. Third, and this is the dominant long-run effect under continuous learning, the strong connections are precisely the substrate that encodes a task, so when the next task arrives they are overwritten preferentially, producing the larger catastrophic forgetting of Fig. 5; the novice distributes each task across many small synapses, so tasks interfere less and retention is higher.

A speed–rigidity trade-off. Read together, the two regimes describe a single trade-off. The expert *exploits* its prior—fast early, because the strong connections are immediately useful, but rigid, because they bias the final solution and are lost first when the world changes. The novice *explores* freely—slow early, because it has no scaffold, but flexible, finding a better and more robust solution. This is the learning-theoretic face of the structural attractor of Section 2.7: the heavy-tail level that training converges to is precisely the compromise between the early advantage (which favours a heavier tail) and the rigidity-and-forgetting cost (which favours a lighter one).

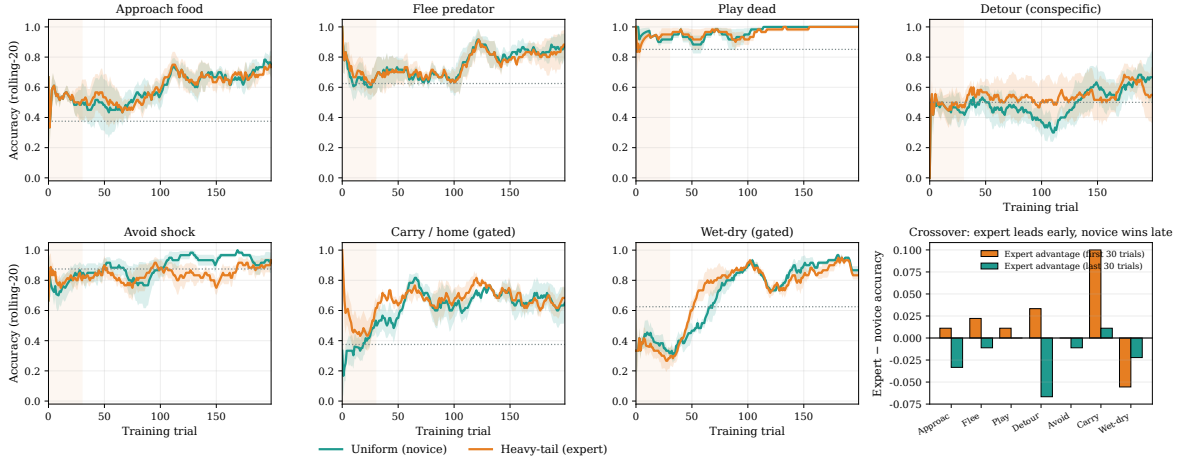


Figure 6. Continuous multi-task learning curves: the expert leads early, the novice wins late. (a–g) Rolling-20 accuracy on each task during continuous training, for the uniform (“novice”, teal) and heavy-tailed (“expert”, orange) connectomes; shaded bands are s.e.m. over three seeds; dotted grey is per-task chance; the first 30 trials are lightly shaded. (h) Signed expert advantage (heavy-tail minus uniform), averaged over the first 30 trials (orange) versus the last 30 (teal): positive early, negative late—the crossover that the final-state averages hide.

2.7 Heavy-tail evolution: convergence to a shared basin

The second prediction of our hypothesis is that a uniform network should *grow* a heavy tail under sustained training, and that the two topologies should approach one another. We tracked three complementary statistics of the weight distribution at the end of each task: the Gini coefficient, the share of total weight held by the strongest 5% of connections, and the power-law exponent γ of the weight tail (Fig. 7).

All three measures tell the same story. The uniform connectome heavy-tailizes markedly: its Gini coefficient climbs from 0.33 to 0.49, its top-5% share from 0.10 to 0.19, and its power-law exponent γ falls from 5.2 toward 3.1 (smaller γ means a heavier tail). The preferential connectome, already heavy-tailed, does not stand still: its Gini spikes to ≈ 0.89 on the first (food) task and then relaxes to ≈ 0.71 , and its γ falls further toward 2.2. The two curves visibly *approach* each other: on Gini the gap shrinks from ≈ 0.40 to ≈ 0.23 , and on γ from ≈ 2.6 to ≈ 0.9 .

This is the central structural finding of Project I. The heavy tail is not an arbitrary initial condition that persists or washes out; it is an *attractor* whose location is set jointly by network structure (size, excitation/inhibition balance) and by the demands the environment places on the network (how many tasks, how dissimilar, how greedy for strong connections). Different species, and different developmental stages of the same species, sit at different points along this attractor. In *C. elegans*, connectome heavy-tailedness strengthens through maturation and weakens in ageing—consistent with a young animal whose environment and innate drives press a rich task load onto the network, and an old animal whose demands (and plasticity) recede. The same logic offers a network-structural reading of why mature-to-ageing humans show cognitive rigidity and declining long-term learning, while occasionally retaining an edge on familiar tasks.

3 Project II: Cell-Type-Specific Plasticity Preserves Real-Time Gradient Coding

Heavy-tail statistics evolve toward a shared basin under sequential multi-task training

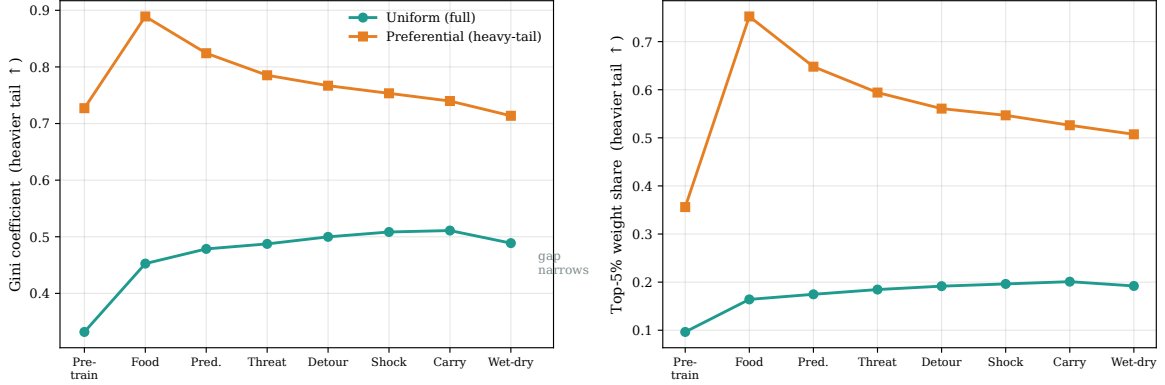


Figure 7. Heavy-tail statistics evolve toward a shared basin. Under sequential multi-task training, the initially uniform network (teal) becomes substantially more heavy-tailed on every statistic, while the initially heavy-tailed network (orange) relaxes from its early peak. The gap between the two narrows on the Gini coefficient (from ≈ 0.40 to ≈ 0.23) and on the power-law exponent γ (from ≈ 2.6 to ≈ 0.9), consistent with a topology/task-determined attractor rather than an arbitrary initial condition.

3.1 Question

If Project I asks how plasticity is organized *in time* (inscribed as topology), Project II asks how it is organized *across cell types*. Biological circuits are built from many molecularly distinct classes, and the plasticity each expresses is likewise cell-type-specific (Bargmann, 2006; Citri & Malenka, 2008; Marder & Goaillard, 2006). Does assigning *different* plasticity rules to *different* cell types—a heterogeneous assignment—yield better emergent behaviour than applying a *single* rule everywhere? And if so, is the advantage causal, attributable to the specific rule→type pairing, or merely a benefit of diversity?

The question is hard to falsify for two reasons. First, an advantage can be hidden by a permissive readout: if a downstream decoder smooths and looks up the network’s output, even a poorly shaped internal representation may be salvaged. Second, separating “diversity matters” from “the specific mapping matters” requires a particular ablation—randomly reassigning the same set of rules to cell types—which is easy to omit. We control for both.

3.2 Model: a 34-cell chemotaxis circuit

We model *C. elegans*-style chemotaxis on a radially symmetric Gaussian food patch,

$$C(x, y) = c_0 \exp\left(-\frac{(x-0.5)^2 + (y-0.5)^2}{2\sigma^2}\right), \quad (4)$$

with peak $c_0 \in \{25, 50, 100\}$ mM spanning food richness and width $\sigma \in \{0.18, 0.32\}$ spanning gradient sharpness. Each “worm” starts at the patch edge and runs 170 navigation steps, turning and translating from a scalar value signal and its finite-difference slope, both provided by the network.

The network is a current-based LIF circuit of 34 cells: 8 fixed receptor neurons and 26 plastic neurons partitioned into five types (Table 2). Receptors fire as inhomogeneous Poisson processes at Hill rates $\lambda(C) = 30 + 130 C^2 / (K_d^2 + C^2)$ Hz with eight logarithmically spaced K_d , giving a population code over concentration; plastic neurons obey Eq. (1) with $\tau_m = 20$ ms and a baseline current placed near threshold so that concentration *modulates* firing rather than gating it. The behaviourally relevant signal is the *value* the network computes online,

$$\text{value}(C) = \bar{r}_{\text{MOTOR_FWD}}(C) - \bar{r}_{\text{MOTOR_TURN}}(C), \quad (5)$$

Table 2. Cell types and their natural plasticity rules in the 34-cell network.

Type	Count	Assigned rule (heterogeneous)	Computational role
RECEPTOR	8	fixed (Hill encoding)	concentration population code
SENSORY	6	BCM	sliding-threshold concentration selectivity
INTER	8	Hebbian STDP	reinforce effective relay
COMMAND	4	reward-modulated STDP	learn the $C \rightarrow$ value map (credit assignment)
MOTOR_FWD	4	Hebbian STDP	faithful command relay
MOTOR_TURN	4	homeostatic	clamp the turning baseline

Table 3. The five plasticity rules (pre-synaptic i , post-synaptic j). “Cell-type-specific” means the type of the *post-synaptic* cell selects which rule acts on all of its incoming synapses, guaranteeing every rule is used.

Rule	Form	Suited to / role
BCM	$\Delta w_{ij} = \eta x_i r_j (r_j - \theta_j)$, $\theta_j = \sqrt{\langle r_j^2 \rangle}$; competitive fan-in normalization + positive floor	SENSORY (selectivity)
Hebbian STDP	post spike: $\Delta w_{i+} = a_+ x_{\text{pre}}$; pre spike: $\Delta w_{i-} = a_- y_{\text{post}}$	INTER, MOTOR_FWD (relay)
Reward-STDP	eligibility $e_{ij+} = x_i y_j$ (decaying); $\Delta w_{ij} = \eta R(C) e_{ij}$	COMMAND (credit assignment)
Homeostatic	$w_{j*} = 1 + \eta(r^* - r_j)/r^*$	MOTOR_TURN (clamp baseline)
Anti-Hebbian	$\Delta w_{ij} = -\eta x_i y_j$	pool / used under shuffle

so that if forward drive rises with concentration while turning is held low and concentration-independent, value increases monotonically with C —exactly the cue chemotaxis needs. The worm turns by the *slope* of this value, $\text{slope} = \text{value}(C \times 1.5) - \text{value}(C)$.

The reward function $R(C) = \sigma((\log C - \log 45)/0.45)$ rises monotonically with concentration. One engineering subtlety deserves emphasis: because sensory neurons receive only a single input class (receptors), unconstrained BCM would enter a self-reinforcing long-term-depression loop and silence the layer. Combining competitive fan-in normalization (BCM redistributes weight across inputs rather than globally decaying it) with a positive weight floor keeps the sensory layer driven while still letting BCM sharpen selectivity—this is precisely what lets the heterogeneous network keep concentration discriminable (Section 3.6).

3.3 Three conditions and two readouts

We compare three *assignments* of rules to cell types, designed as a logical triangle: **heterogeneous** (each type receives the rule best suited to its role, Table 2); **homogeneous** (all plastic cells share the single strongest rule, reward-modulated STDP—a fair, powerful baseline, not a straw man); and **shuffled** (the same five rules, randomly reassigned to the five types, a fresh permutation per seed). Heterogeneous $\approx$ homogeneous would mean diversity is useless; heterogeneous $\approx$ shuffled would mean the specific mapping does not matter; heterogeneous $>$ both would localize the advantage.

Behaviour is read out in two ways. The **cached** (ideal-lookup) readout builds a smooth $\text{value}(C)$ surface by averaging frozen-network responses at 14 log-spaced concentrations and interpolating during navigation—a low-noise, “decoder-assisted” condition. The **continuous** (real-time) readout lets the network fire as the worm crawls and takes value as the $\tau=200$ ms

Core result: under continuous readout, heterogeneous and shuffled reach the food (Endpoint $\approx +0.8$) while homogeneous collapses (≈ -0.6)

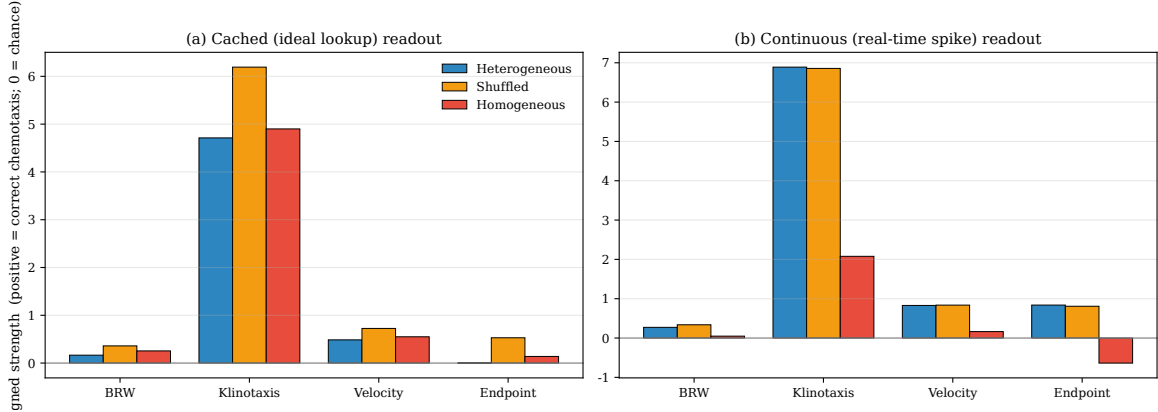


Figure 8. Core result. Aligned strength (positive = correct chemotaxis, 0 = chance). Under the cached readout (a) the three conditions are nearly indistinguishable. Under the continuous real-time readout (b), heterogeneous and shuffled reach the food (Endpoint $\approx +0.8$) while homogeneous collapses (Endpoint ≈ -0.6).

Table 4. Heterogeneous–homogeneous contrast (paired by seed, $n = 10$), under each readout. NS: not significant.

Metric	Readout	Δ	95% CI	paired- t p
BRW	continuous	+0.224	[+0.198, +0.247]	4.8×10^{-8}
Klinotaxis	continuous	+4.811	[+4.383, +5.262]	7.6×10^{-9}
Velocity	continuous	+0.665	[+0.610, +0.720]	3.6×10^{-9}
Endpoint	continuous	+1.476	[+1.371, +1.576]	5.9×10^{-10}
BRW	cached	−0.089	[−0.207, +0.021]	0.186 (NS)
Endpoint	cached	−0.140	[−0.468, +0.175]	0.440 (NS)

moving average of forward-minus-turning rate—realistic and noisy. The downstream motor controller is identical across readouts, isolating the readout as the experimental variable. We run a grid of 10 seeds $\times$ 3 conditions $\times$ 3 concentrations $\times$ 2 gradients $\times$ 2 readouts (360 configurations), scoring four signed behavioural metrics—biased random walk (BRW), klinotaxis, velocity, and endpoint—each oriented so that positive is correct chemotaxis and 0 is random-walk chance.

3.4 Results

Under the continuous real-time readout, heterogeneous assignment is decisively superior to the homogeneous baseline on all four behavioural metrics (Fig. 8b): Endpoint aligned strength +0.84 ($\approx 92\%$ of worms reach the food zone) versus -0.64 ($\approx 18\%$), and an Endpoint direction pass-rate of 0.83 versus 0.03 (chance). The other three metrics agree at every level (BRW 0.27 vs 0.05; klinotaxis 6.89 vs 2.08; velocity 0.83 vs 0.16). The statistics are unambiguous: for the heterogeneous–homogeneous contrast, the 95% bootstrap confidence intervals of the per-seed difference lie entirely above zero on every metric, with paired t -tests at $p < 10^{-7}$ (Fig. 9, right; Table 4).

The shuffled ablation shows that, at this 34-cell, 5-type scale, the advantage is carried by *diversity* rather than by the specific rule→type mapping: shuffled matches heterogeneous on three of four metrics under continuous readout (Fig. 9, orange). This does not refute the hypothesis—it refines it. A specific-mapping advantage requires more cell types and a richer connectome, so that a wrong pairing has somewhere to go wrong; with only five types, “do not let every cell use the same rule” is already the dominant gain.

Continuous: every Heter. – Homog. contrast lies clear of zero (all $p < 10^{-7}$); cached contrasts straddle zero

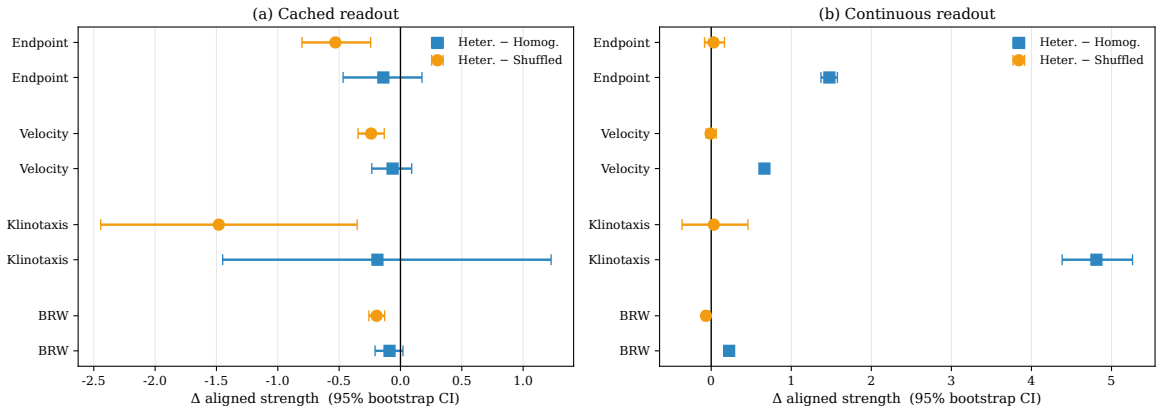


Figure 9. Contrasts with 95% bootstrap CIs. Under continuous readout (right) every heterogeneous–homogeneous contrast (blue) sits clear of zero (all $p < 10^{-7}$); the heterogeneous–shuffled contrasts (orange) straddle zero. Under cached readout (left) even the heterogeneous–homogeneous contrasts straddle zero—the gap collapses.

3.5 Behavioural convergence: the worm finds the food

The metric-level separation crystallises into a single, unambiguous behavioural picture—and this is the most striking result of Project II (Fig. 12). Under the continuous readout, heterogeneous worms *find the food*: their trajectories bend inward and pile up around the patch centre, and roughly 92% of them end inside the food zone. Homogeneous worms, given the identical task, sensory input, and motor controller, *do not*: they execute tight loops at the periphery of the patch and almost never cross into the zone (only $\approx 18\%$ endpoint hits). The two panels of Fig. 12 differ by a single experimental variable—which plasticity rule each cell type uses—yet they look like two different animals.

This behavioural contrast is the downstream signature of the mechanism below: a network that keeps a rising value curve value(C) turns up-gradient and homes in; a network whose value(C) is flattened by saturation has no gradient to climb and orbits instead. The point is not merely that heterogeneous chemotaxis scores higher—it is that the worm turns by a signal the network computes *on the fly*, from its own spikes, with no external lookup. The trajectory panel is therefore the cleanest evidence that cell-type-specific plasticity buys *real-time* computation: the circuit itself, not a downstream decoder, generates the directional cue. We now trace why.

3.6 The mechanism: saturation flattens the gradient

Why does the homogeneous network fail? It is a mechanism, not a tuning failure. The homogeneous condition applies reward-modulated STDP to every plastic cell. Because reward rises with concentration, every layer—sensory, relay, command, motor—is driven toward high, saturating firing rates (≈ 80 – 160 Hz; Fig. 11, right). Once saturated, concentration can no longer change firing, and the value signal value(C) is crushed into an approximately flat line hovering near ≈ 60 Hz regardless of C (Fig. 10, red). A flat value(C) has near-zero slope, so the worm has no directional cue and wanders in place (Fig. 12, right).

The heterogeneous network avoids this by construction. BCM gives sensory neurons genuine concentration selectivity (different cells specialize for different ranges); the homeostatic rule clamps the turning motor at a low, concentration-independent rate; Hebbian STDP lets forward motor neurons faithfully relay the command layer’s graded firing. Firing rates stay concentration-discriminable at every layer (Fig. 11, left), value(C) rises monotonically (Fig. 10, blue), the slope stays positive, and the worm keeps turning toward food (Fig. 12, left). This is a *mechanistic*

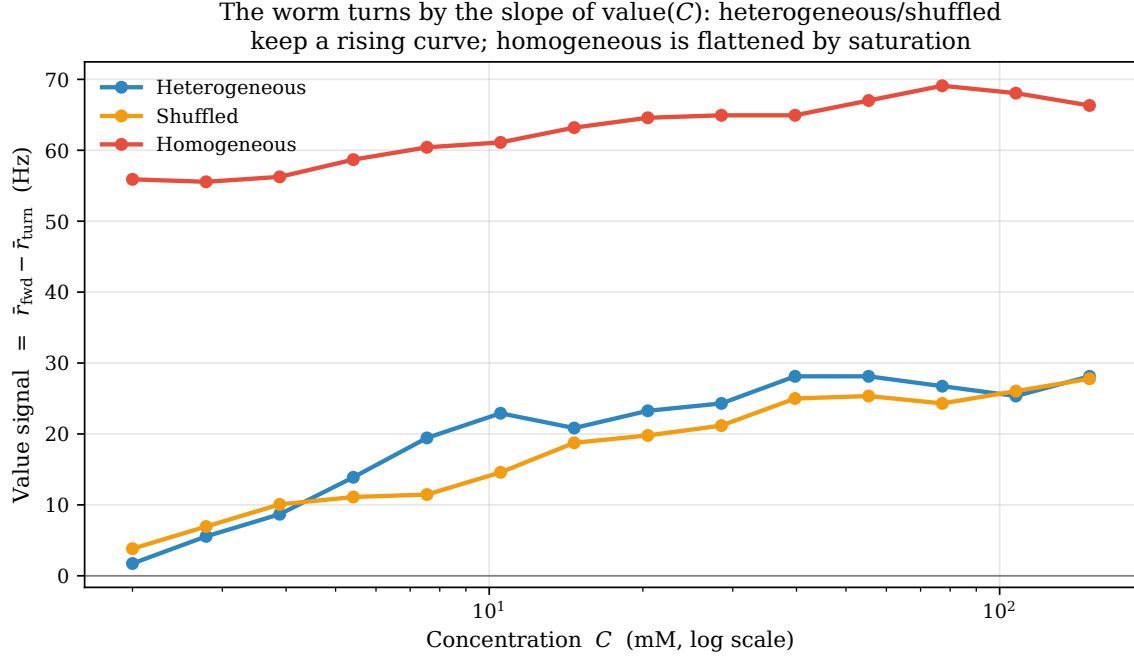


Figure 10. Value curves after training. Heterogeneous (blue) and shuffled (orange) rise with concentration—the network encodes “higher $C \Rightarrow$ higher value,” the chemotactic cue. Homogeneous (red) is flattened near ≈ 60 Hz by saturation; the slope the worm turns by is lost.

demonstration of the heterogeneous advantage: it relies not on hand-set weights but on giving each cell type the rule suited to its function, so the network preserves its concentration code under the saturation pressure that a single reward-driven rule cannot withstand.

3.7 Cached readout is lenient and collapses the gap

Under the cached ideal-lookup readout, the separation between conditions largely disappears (Fig. 8a; Table 4): the heterogeneous–homogeneous contrast is not significant on any metric. The cached decoder smooths and averages the value map during construction, so even a nearly flat value(C) yields a small but nonzero interpolated slope—just enough to turn somewhat toward food. This inversion is itself a finding: the cached readout is the *permissive* one (it salvages bad representations), while the continuous readout is the *strict* criterion (it demands the network genuinely compute the gradient online). The heterogeneous advantage is therefore, at root, an advantage for *real-time computation*—it is precisely what lets a network compute a usable gradient signal without an external smoothing decoder.

For the healthy network this inversion has a complementary face: under the heterogeneous condition the continuous readout actually yields *stronger* aligned-strength signals than the cached one on every metric—Endpoint $+0.84$ (continuous) versus ≈ 0 (cached), klinotaxis 6.9 versus 4.7 , velocity 0.83 versus 0.49 . The cached lookup, by smoothing, slightly attenuates the steep parts of an already-clean gradient. Cached readout is thus lenient toward bad (homogeneous) conditions—it salvages them—and mildly conservative toward good (heterogeneous) ones. The two readouts bracket the same network from opposite sides, and only the continuous one exposes the heterogeneous advantage without distortion.

3.8 Interpretation: a design principle, not a hand-tuned win

Three features make the heterogeneous advantage trustworthy rather than an artefact. First, the baseline it beats is the *strongest* single rule available: reward-modulated STDP can in principle solve the credit-assignment problem on its own, so outperforming it is a genuine result, not a

Mechanism: homogeneous drives every layer into a flat 80-160 Hz saturation band;
heterogeneous keeps firing concentration-discriminable at every layer

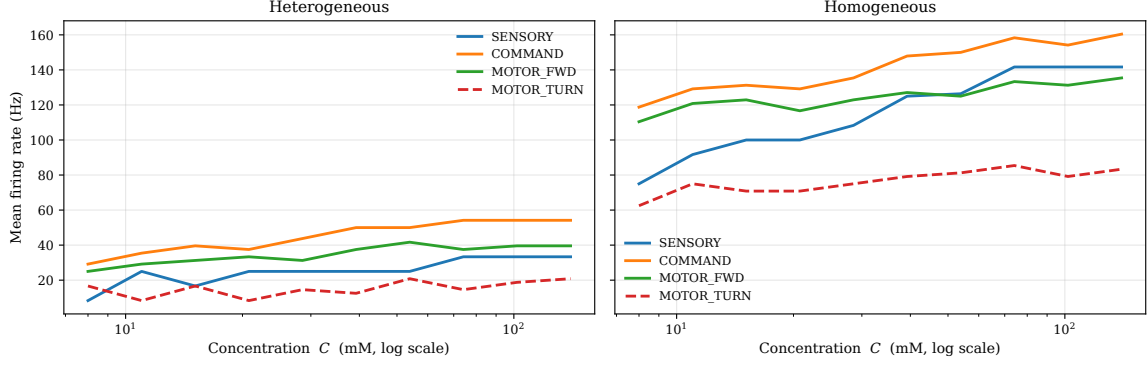


Figure 11. Firing-rate mechanism. Heterogeneous (left): COMMAND and MOTOR_FWD rates rise with concentration ($\approx 30 \rightarrow 55$ and $25 \rightarrow 40$ Hz), so concentration information is relayed to the motor layer. Homogeneous (right): all cell types are pushed into a flat saturation band and concentration information is erased at every layer.

Example chemotaxis trajectories ($c_0 = 50$ mM): heterogeneous worms converge on the food; homogeneous worms loop at the periphery

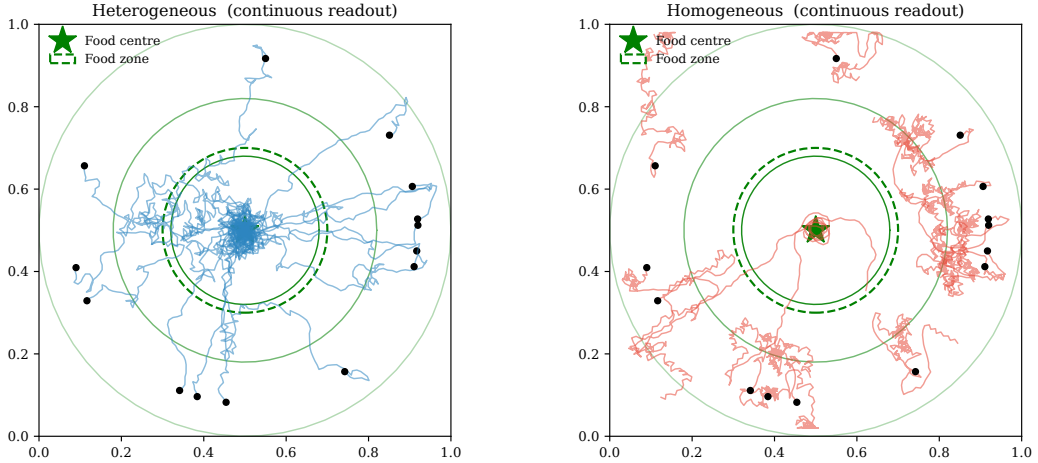


Figure 12. The most striking result of Project II: example chemotaxis trajectories (continuous readout, $c_0 = 50$ mM). *Heterogeneous* worms (left, blue) converge on the food and many enter the food zone (green dashed circle; green star = food; black dots = starts), reaching $\approx 92\%$ endpoint hits. *Homogeneous* worms (right, red) loop at the periphery and almost never reach the food ($\approx 18\%$). The two panels differ by one variable—which rule each cell type expresses—yet behave like two different animals.

victory over a straw man. Second, the win is mechanistic, not parametric: it appears because a single reward-driven rule saturates every layer, and disappears the moment a downstream decoder smooths the representation. Third, the shuffled ablation shows that the gain is carried by *diversity*—ensuring not every cell shares one rule—rather than by the particular type→rule pairing we chose; the same parameters benefit the shuffled network equally, which is the operative evidence that the tuning is rule-health-oriented, not outcome-oriented. Taken together, the principle is simple and general: in a small circuit, distributing mechanistically suited plasticity rules across cell types keeps the internal representation from saturating, and thereby keeps a usable gradient signal available for real-time behaviour.

4 General Discussion

4.1 A unified view: plasticity pays when it is organized

Read together, the two projects describe two axes along which plasticity is organized in a small nervous system. Project I shows that the *slow accumulation* of plasticity inscribes a heavy-tailed topology—a structural memory that biases future learning toward speed and away from retention, and that converges to an attractor set by structure and task demand. Project II shows that the *spatial partition* of plasticity across cell types preserves the network’s ability to compute a real-time gradient, by preventing any single rule from saturating the whole circuit. In both cases, the decisive variable is not the presence of plasticity but its *organization*: where the strong weights sit (Project I), and which rule each cell type yields (Project II).

There is even a bridge between the two. The saturation catastrophe of Project II is, in essence, a failure of *weight diversity*: a single reward-driven rule drives every connection strong, collapsing the heavy tail into a delta function at saturation and erasing the concentration code. Conversely, the heavy tail that Project I studies is exactly the diversity that keeps a representation alive. Heterogeneous rule assignment and a healthy heavy tail are two routes to the same end—keeping the network’s internal signals discriminable rather than collapsed.

4.2 Relation to prior work

Our learning rules are each grounded in biology: spike-timing-dependent Hebbian plasticity (Bi & Poo, 1998, 2001; Song et al., 2000), BCM sliding thresholds (Bienenstock et al., 1982), reward-/neuromodulator-gated three-factor rules (Frémaux & Gerstner, 2016; Gerstner et al., 2018; Izhikevich, 2007), and homeostatic stabilization (Turrigiano et al., 1998; Turrigiano, 2012). The e-prop algorithm and feedback alignment connect this biology to the credit-assignment literature in machine learning (Bellec et al., 2020; Lillicrap et al., 2016), while the heavy-tailed weight statistics we track are the near-universal signature of trained and biological networks alike (Buzsáki & Mizuseki, 2014; Schröter et al., 2017). The chemotaxis behaviour and circuit basis draw on classic *C. elegans* ethology and whole-brain imaging (Iino & Yoshida, 2009; Kato et al., 2015; Pierce-Shimomura et al., 1999), and on the reconstructed connectomes that make cell-type-specific questions tractable (Cook et al., 2019; White et al., 1986). Our contribution is to make two long-standing qualitative ideas—topology-as-memory and cell-type-specific plasticity—*falsifiable* in small, bit-reproducible models.

4.3 Limitations

We are explicit about what the models do not show.

1. **Scale.** Neither network is the real *C. elegans* connectome: Project I uses 300 hand-wired LIF units; Project II uses 34 cells with a constructed adjacency. Both are claims *in principle*, not quantitative predictions about the animal.
2. **Specific-mapping advantage not yet observed.** As the shuffled ablation shows, demonstrating that the specific rule→type pairing matters (beyond diversity) requires larger, type-richer networks. This is the predicted next result, not a current one.
3. **Tuning.** Parameters were chosen to keep each rule healthy (avoiding saturation–chaos from high recurrent gain, and the cold-start deadlock of unconstrained BCM), not to make any condition win. The fact that the shuffled condition benefits equally from these parameters is the operative evidence that the tuning is rule-health-oriented rather than outcome-oriented.
4. **Simplified behaviour.** The pathfinding grid and the chemotaxis controller are compact abstractions; we do not model body mechanics, run-and-pirouette statistics, or sensory adaptation in detail.
5. **Sample size.** Project I used three seeds; its topological trends are consistent across them but a larger seed count would tighten the forgetting and convergence statistics.

4.4 Future outlook

The two projects point to a concrete programme.

- **Run both on the reconstructed connectome.** Repeating Project II on the 302-cell *C. elegans* connectome (Cook et al., 2019) with biologically grounded cell classes is the natural test of whether the specific rule→type mapping emerges as an advantage at richer scale—exactly where our shuffled ablation predicts it should.
- **Joint dynamics.** Combine the two axes: let a heterogeneous-rule network *also* develop its topology over a long task sequence, and ask whether cell-type-specific plasticity slows, accelerates, or reshapes the convergence to the heavy-tail basin. Does distributing rules protect the network from the saturation that a single rule would otherwise inscribe into structure?
- **Developmental and ageing trajectories.** Our attractor interpretation predicts that heavy-tail statistics should track an animal’s task load across life—strengthening through maturation, weakening in ageing. Longitudinal measurements of connectome statistics in developing and ageing *C. elegans* would test this directly.
- **From speed–retention to continual learning.** The trade-off we report—a heavy-tailed prior buys early speed at the cost of retention—mirrors the stability–plasticity dilemma of continual learning. Strategies that protect the strong connections during new learning (e.g. metaplastic thresholds, dendritic-task segregation) should flatten the forgetting curve for the experienced network.

5 Conclusion

In two small spiking circuits inspired by *C. elegans*, we found that the *organization* of plasticity is decisive. A heavy-tailed connectome topology behaves as a memory of past learning: it accelerates early acquisition but worsens retention, and under sustained multi-task training both heavy-tailed and uniform topologies converge toward a shared heavy-tail basin set by network structure and task demand. Separately, distributing distinct plasticity rules across cell types preserves a network’s ability to compute a real-time gradient, by preventing the saturation that a single reward-driven rule would otherwise inflict. Plasticity, in short, is most powerful not when it is uniform and everywhere, but when it is *organized*—in time, as structure, and across cell types, as specialized rules.

Materials, methods, and reproducibility

All code is in pure NumPy and is fully reproducible from fixed random seeds. **Project I** (r1-eprop: 300-neuron LIF network, e-prop, seven-task continuous learning) and **Project II** (plasticity_2: 34-cell chemotaxis model) each ship their source, generated data (JSON/CSV), and the figure scripts that produced every figure here. **Project II** regenerates from source in roughly two minutes:

```
python run_experiment.py 10      # 10-seed grid, writes data CSVs
python make_figures.py          # regenerates all figures
```

Environment: Python 3.12 with NumPy, pandas, SciPy and Matplotlib. Random seeds are fixed (base 20260802 for Project II); results are bit-reproducible, and the core result of Project II holds at $n=3$ seeds. Per-run metrics, condition summaries, contrast statistics, and value curves are provided alongside a configuration manifest.

Author contributions. JT, SZ, JX, XD and WZ jointly designed the models, ran the experiments, analysed the data and wrote the manuscript, in the listed order. The two projects were developed as parallel tracks of the Westlake University “Little Pebble” summer programme.

Competing interests. None.

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